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30 June 1981
SUPERSEDING
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1 July 1976

MILITARY SPECIFICATION

RANGEFINDER, FIRE CONTROL, LASER: 11743101

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE OBSERVED IN OPERATING AND TESTING THE RECEIVER-TRANSMITTER. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE AND BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL AND PAINT. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE, THE RECEIVER-TRANSMITTER SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Rangefinder, Fire Control, Laser: 11743101.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411 Nitrogen, Technical

FSC 1240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration Systems Requirements

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11743101	Rangefinder, Finder Control, Laser: AN/VVG-2

Inspection Equipment

11750222	Test Set - Laser System Test
11750232	Cabinet, Laser - F.A.T.
11821748	Test Set - Immersion Leak, Purge and Fill

Packaging Data Sheet

P11743101	Packaging of Rangefinder, Fire Control, Laser

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The laser fire control rangefinder, herein referred to as the rangefinder, shall be manufactured in accordance with Drawing 11743101 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The rangefinder shall transmit a pulse of laser light, receive reflected light from the target and convert the time interval from transmission to reception into range data.

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	$73^{o} \pm 18^{o}F$
b.	Relative humidity	50 percent $\pm$ 30 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the rangefinder shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3. The pulse forming network (PFN) current may be adjusted to meet the rangefinder performance requirements but shall not be adjusted further from the initial setting in order to meet the environmental requirements. The rangefinder shall operate as specified herein with the power and signals of table I applied as specified.

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TABLE I. Power and signals.

Item	Condition	Characteristics	Electronic Unit Connections
1.0	Power Source		Connected between the following terminals:
1.1	Tank Power	24 ± 6 Vdc Ripple: 200 mV p-p max	J4-A and B (+) and J4-C and D (-)
1.2	Tank Power	24 ± 6 Vdc Ripple: 200 mV p-p max	Applied between J6-A (+) and J6-L (-) when specified herein.
1.3	Plus 15 Vdc	15.0 ± 0.7 Vdc Ripple: 0.5 V p-p max	Applied between J6-J (+) and J6-L (-)
1.4	Minus 15 Vdc	-15.0 ± 0.7 Vdc Ripple: 0.5 V p-p max	Applied between J6-K (-) and J6-L (+)
1.5	AC Ref 1	5.000 ± 0.010 Vrms 400 ± 4 Hz	Applied between J6-M and J6-L
1.6	AC Ref 2	5.000 ± 0.010 Vrms 400 ± 4 Hz, also must be out-of-phase with AC Ref 1 by 180° ± 2.25°	Applied between J6-N and J6-S
1.7	115V, 400 Hz	115.0 ± 5.0 Vrms 400 ± 40 Hz	Applied between J6-P and J6-R when specified herein
2.0	Signal Source		
2.1	Remote reset/ range	5.0 ± 0.5 Vdc	Applied between J7-R (+) and J7-N (-) when specified herein
2.2	Analog 0 to 36 V	0 to 36 Vrms, 400 ± 40 Hz, must be in-phase or out-of-phase with 115V, 400 Hz source of item 1.7	Applied between J6-W and J6-X when specified here-in

3.3.1 Power turn-on condition. With the EMER POWER switch set to the ON position, the following conditions shall exist at the commander's control panel: (See 4.6.2.1)

a. The RANGE and RETURNS shall indicate zero.

b. The RESET, FEED, LAST and BATTLE RANGE lamps shall illuminate.

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c. One of the MODE lamps shall illuminate and correspond to the position of the MODE switch.

d. The RANGE lamp shall illuminate steady or flash and correspond to the position of the MODE selector switch.

e. With the EMER POWER switch set to the OFF position after the power turn-on conditions are met, the analog voltage between J6-D and J6-L shall be 1.70 ± 0.25 Vrms.

3.3.2 Ranging operation. The RANGE button shall function in accordance with requirements specified in table II and in the cyclical numerical sequence established by the item numbers of that table. (See 4.6.2.2)

TABLE II. Ranging operation.

Item	RANGE button lamp	Button	Action	RESULT		
				PFN	RANGE button lamp	LAST lamp
1	Steady	RANGE	Pressed	No change	No change	No change
2	Steady	RESET	Pressed	No change	Flashing	Steady
3	Flashing	RANGE	Pressed	Discharges	Out	Out
4	Out	RANGE	None	Discharging	No change	No change
5	Steady	RANGE	None	Charged	Steady	No change

3.3.3 Range return selector. With the SELECT indicator lamp illuminated and with buttons 1, 2 and LAST pressed singly in any sequence, each range of not more than the first three target returns shall be displayed singly in the same sequence. The consecutive button identification shall correspond to the first, second and third target returns. (See 4.6.2.3)

3.3.4 RESET switch. With the RESET button pressed, the RANGE and RETURNS readouts shall be zero. (See 4.6.2.4)

3.3.5 BATTLE RANGE switch. With the MODE switch in the ON, TEST, or AUTO position and with the BATTLE RANGE switch pressed, the 1200 meter range shall be generated and strobed into the computer, the RANGE and RETURNS readouts shall be zero and the BATTLE RANGE lamp shall illuminate. The analog voltage between J6-D and J6-L shall be 1.277 ± 0.10 Vrms. (See 4.6.2.5)

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3.3.6 Laser MODE switch. With the laser MODE switch positioned to TEST, ON or AUTO, the respective mode lamp shall illuminate. (See 4.6.2.6)

3.3.6.1 TEST position. With the MODE switch in the TEST position, the RANGE lamp shall illuminate steady and the PFN shall not discharge through the flashtube with the RANGE button pressed.

3.3.6.2 ON position. With the MODE switch in the ON position, the RANGE lamp shall flash. With ranging performed, the range to target shall be displayed on the RANGE readouts and the SELECT lamp shall illuminate. With the FEED switch pressed, the SELECT lamp shall extinguish and the GO lamp shall illuminate.

3.3.6.3 AUTO position. With the MODE switch in the AUTO position, the RANGE lamp shall flash. With ranging performed, the range to the target shall be displayed on the RANGE readouts and the GO lamp shall illuminate.

3.3.7 Ranging limits, accuracy, resolution and rate. Ranging shall conform to the following: (See 4.6.2.7)

a. The range shall be not less than 200 meters.

b. The range shall be not greater than 5000 meters.

c. The range accuracy shall be plus or minus 10 meters.

d. The range resolution shall be 20 meters.

e. Minimum time between ranging: With the RANGE button pressed, the flashing red lamp on the RANGE button shall extinguish and illuminate steady in not greater than 4 seconds.

f. The maximum number of rangings shall be either 6 rangings per minute for 2 minutes or 3 rangings per minute sustained without deterioration of range accuracy.

3.3.8 Light dimming control. With rotation of the LIGHT control knob in a clockwise (CW) direction from the DIM position, the brightness of all lamps except the RANGE and RETURNS readouts shall increase in brightness. Counterclockwise (CCW) rotation shall decrease the brightness. (See 4.6.2.8)

3.3.9 LIGHT switch. At the extreme CW position of the LIGHT control shall be a test switch for the lamps. With rotation of the LIGHT control to the extreme CW position, all the lamps on the control panel shall illuminate and eights shall register on the RANGE and RETURNS readouts. (See 4.6.2.9)

3.3.10 Peak output power. The laser peak output power of the range finder shall be not less than one megawatt. (See 4.6.2.10)

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3.3.11 Input voltage variation. The rangefinder shall meet the ranging accuracy specified in 3.3.7c with the input voltage of the prime source varied between 18 Vdc and 30 Vdc. (See 4.6.2.11)

3.3.12 Power drain. The direct current drawn from the prime power source shall be not greater than 12 amperes. (See 4.6.2.12)

3.3.13 Primary power transient. The RANGE readout shall be unchanged under the following conditions: (See 4.6.2.13)

a. With the input voltage dropped to 9.0 + 1.0 -0.0 volts for a duration of 100 ± 10 milliseconds.

b. With the input voltage increased to 90 + 0 -5 volts for a duration of 100 ± 10 milliseconds.

3.3.14 Transmitter test. With the EMER POWER switch in the XMTR TEST position and with the rangefinder fired, there shall be no MALF 3 indication. (See 4.6.2.14)

3.3.15 Built-in test (BIT). The rangefinder shall be capable of self test in accordance with table III. (See 4.6.2.15)

3.3.16 Remote controlled operations. (See 4.6.2.16)

3.3.16.1 Remote power - ON. With the voltage of item 1.2 of table I applied, the rangefinder shall be operational.

TABLE III. Controls and indicators test.

Item	Inputs 3/		Indications				
	Press and Release Switches as Indicated	Range Return Selector Lamp	Returns Readout	Range Readout ±15	GO Lamp (Green)	SELECT Lamp (Yellow)	Range Return Selector Locked Up
1	RESET	LAST	0	0000	OFF	OFF	-
2	RANGE	LAST	1	850	ON	OFF	YES 1/
3	RANGE	LAST	2	1850	ON	OFF	-
4	RANGE	LAST	3	2850	ON	OFF	-
5	RANGE	LAST	4	2850	OFF	ON	NO 1/
6	1	1	4	850	OFF	ON	-

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TABLE III. Controls and indicators test. - Continued

Item	Inputs 3/		Indications				
	Press and Release Switches as Indicated	Range Return Selector Lamp	Returns Readout	Range Readout ±15	GO Lamp (Green)	SELECT Lamp (Yellow)	Range Return Selector Locked Up
7	2	2	4	1850	OFF	ON	-
8	FEED	2	4	1850	ON	OFF	YES 1/
9	RESET	LAST	0	0000	OFF	OFF	-
10	2	2	0	0000	OFF	OFF	-
11	RANGE	2	1	9995 2/	OFF	ON	-
12	FEED	2	1	9995	OFF	ON	-
13	RANGE	2	2	1850	OFF	ON	-
14	FEED	2	2	1850	ON	OFF	YES
15	RESET	LAST	0	0000	OFF	OFF	-
16	1	1	0	0000	OFF	OFF	-
17	RANGE	1	1	850	ON	OFF	YES
18	RANGE	1	2	850	ON	OFF	-
19	RANGE	1	3	850	OFF	ON	-
20	RESET	LAST	0	0000	OFF	OFF	-

1/ Verify by pressing 1, 2, and LAST buttons on range return selector.

2/ If Range Readouts read 000 or 9990, reset system and repeat test sequence.

3/ Place MODE control in TEST position.

3.3.16.2 Remote reset and ranging. With the voltage pulse, item 2.1 of table I, applied, the laser shall fire and provide status information. With this voltage pulse reapplied, the laser shall fire and provide status information.

3.3.16.3 Remote BIT. With a ground applied to J6-C with respect to J6-L, the rangefinder shall provide a BIT analog range voltage between J6-D and J6-L of 1.968 ± 0.20 Vrms.

3.3.16.4 Azimuth mirror drive interface. The requirements of 3.3.16.4.1 and 3.3.16.4.2 shall be met with 115 Vrms, 400 Hz, (item 1.7 of table I), applied to J6-P and J6-R after boresight position is achieved. Boresight is that condition in which the line of sight through the reticle is perpendicular

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to the plane through the axis of rotation and the ballistic drive linkage bearing centerline.

3.3.16.4.1 Laser feedback voltage. With the azimuth boresight knob set to 4.0 and 36 ± 1 Vrms at 400 Hz, item 2.2 of table I, applied to J6-W and J6-X, the laser feedback voltage between J6-S and J6-Y shall be not less than 5.94 V peak-to-peak.

3.3.16.4.2 Tachometer output. With 36 ± 1 Vrms at 400 Hz, item 2.2 of table I, applied to J6-W and J6-X, the tachometer voltage between J6-U and J6-V shall be not less than 2.5 Vrms.

3.3.16.4.3 Boresight knob travel - elevation. The elevation boresight knob travel shall be not less than ± 5.0 mils from the boresight position.

3.3.16.4.4 Boresight knob travel - azimuth. With the mirror positioned to achieve the boresight condition and the azimuth boresight knob adjusted for the null condition, 10 mV rms at J6-Y, the azimuth boresight knob shall be further adjustable CW and CCW to obtain a feedback of not less than 295 mV rms, 5.0 mils, at J6-Y. With CW rotation of the boresight knob the feedback voltage shall be in phase with AC Ref 1, (item 1.5, table I), and with CCW rotation the feedback voltage shall be out of phase with AC Ref 1.

3.3.16.4.5 Resolver (mirror) accuracy. From the null position per 3.3.16.4.4, the resolver feedback voltage shall be as specified in table IV.

3.3.17 Sight system. (See 4.6.2.17)

3.3.17.1 Resolution. Resolution on the optical axis shall be not greater than 7 arcseconds at 12X and shall be not greater than 10 arcseconds at 6X. At 0.7 field, the resolution shall be not greater than 15 arcseconds at 12X and shall be not greater than 30 arcseconds at 6X.

TABLE IV. Laser feedback.

Item	Target Position in Mils	Laser Feedback (mV rms) at pins J2-AA and J2-E (rtn)	
1	0	0 ± 10	
2	12 right	707 ± 10	
3	24 right	1414 ± 10	In phase with AC Ref 1
4	36 right	2121 ± 10	
5	0	0 ± 10	
6	12 left	707 ± 10	
7	24 left	1414 ± 10	Out-of-phase with AC Ref 1
8	36 left	2121 ± 10	
9	0	0 ± 10	

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3.3.17.2 Parallax. Parallax at the center of the field shall be not greater than 0.1 mil with the target viewed at a range of 1000 ± 100 meters.

3.3.17.3 Image tilt. The sight system shall rotate the image of an object not greater than two degrees.

3.3.17.4 Parallelism of reticle and image. The image of a line parallel to the axis of rotation of the control, commander's sight and receiver-transmitter, laser ranging assembly shall be parallel to the center horizontal reticle line by not greater than one degree.

3.3.17.5 Eyepiece focus. Eyepiece focus shall be adjustable over a range of plus and minus 2.80 diopters.

3.3.17.6 Reticle illumination. With the reticle lamp control set for normal illumination, the reticle pattern shall be clearly defined in a darkened area. Rotation of the reticle light control CW shall increase the intensity of the reticle illumination.

3.3.18 Optical operability. (See 4.6.2.18)

3.3.18.1 Diopter adjustment torque. The torque required to adjust the diopter setting shall be not less than 3.0 and not greater than 18.0 inch-pounds under the conditions of 3.1.3 and shall be not greater than 20.0 inch-pounds at -25°F and +125°F.

3.3.18.2 Boresight knob torque. The torque required to rotate the boresight knobs shall be not less than 0.5 and not greater than 8.0 inch-pounds under the conditins of 3.1.3 and shall be not greater than 12.0 inch-pounds at -25°F and +125°F.

3.3.18.3 Boresight knob disengagement. The force required to disengage the boresight knobs from their normal positions shall be not less than 5.0 and not greater than 12.0 pounds under the conditions of 3.1.3 and shall be not greater than 18.0 pounds at -25°F and +125°F.

3.3.18.4 Magnification knob torque. The torque required to change magnification shall be not less than 6.0 and not greater than 12.0 inch-pounds under the conditions of 3.1.3 and shall be not greater than 18.0 inch-pounds at -25°F and +125°F.

3.3.18.5 Boresight retention. An established boresight position shall change not greater than ± 0.25 mil in azimuth and ± 0.35 mil in elevation during exposure to the environmental conditions specified in 3.4 except after subjection to the storage temperature conditions, -65°F to +160°F, the change in boresight shall be not greater than ±0.5 mil in azimuth and ±0.7 mil in elevation.

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3.3.19 Sight system and laser beam alignment. (See 4.6.2.19)

3.3.19.1 Initial alignment. The laser transmitted beam shall be parallel to the reticle centerline by not greater than 0.1 mil.

3.3.19.2 Alignment retention. Established alignment of 3.3.19.1 shall vary by not greater than ±0.25 mil during exposure to the environmental conditions specified in 3.4.

3.4 Environmental.

3.4.1 Shock. The rangefinder shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity unit (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the rangefinder shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The rangefinder shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profiles of figure 1 Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The rangefinder shall be capable of operating as specified herein over the operating temperature range of -25°F to +125°F. (See 4.6.3.3 and 4.6.3.4)

3.4.4 Storage temperature. The rangefinder shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to 160°F. (See 4.6.3.3 and 4.6.3.4)

3.4.5 Humidity. The rangefinder shall operate as specified herein at up to 100 percent relative humidity. (See 4.6.3.5)

3.4.6 Altitude. The rangefinder shall operate as specified herein under the following conditions:

a. during exposure to an atmospheric pressure equivalent to 10000 feet above sea level.

b. after exposure to an atmospheric pressure equivalent to 40000 feet above sea level for not less than 1 hour. (See 4.6.3.6)

3.4.7 Dust. The rangefinder shall operate as specified herein after exposure to moving dust for not less than 12 hours at velocities up to 1750 feet per minute. (See 4.6.3.7)

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3.4.8 Salt fog. The rangefinder shall operate as specified herein after exposure to a 5 percent sodium chloride atomized spray for not less than 48 hours. (See 4.6.3.8)

3.4.9 Water hosing. The rangefinder shall withstand water on all surfaces and joints at 50 psi for not less than 30 seconds applied through a 1 inch nozzle at not more than 3 feet distance without intake of water. (See 4.6.3.9)

3.5 Sealing. With an internal pressure of dry nitrogen at 5.0 ± 1.0 pounds per square inch gauge (psig), the rangefinder shall exhibit a pressure drop not greater than 0.2 psig for not less than 2 hours. The optical and transmitter compartments shall be purged with dry nitrogen at 12 ± 1 psig for not less than 30 minutes with a final pressure of 4.5 ± 0.5 psig. All other compartments shall be purged with dry nitrogen at 1 psig for not less than 5 minutes with a final pressure of 0.0 + 0.2 - 0.0 psig. Dry nitrogen shall be BB-N-411, type I, grade B, class 1 with a dew point not greater than -25°F. (See 4.6.5)

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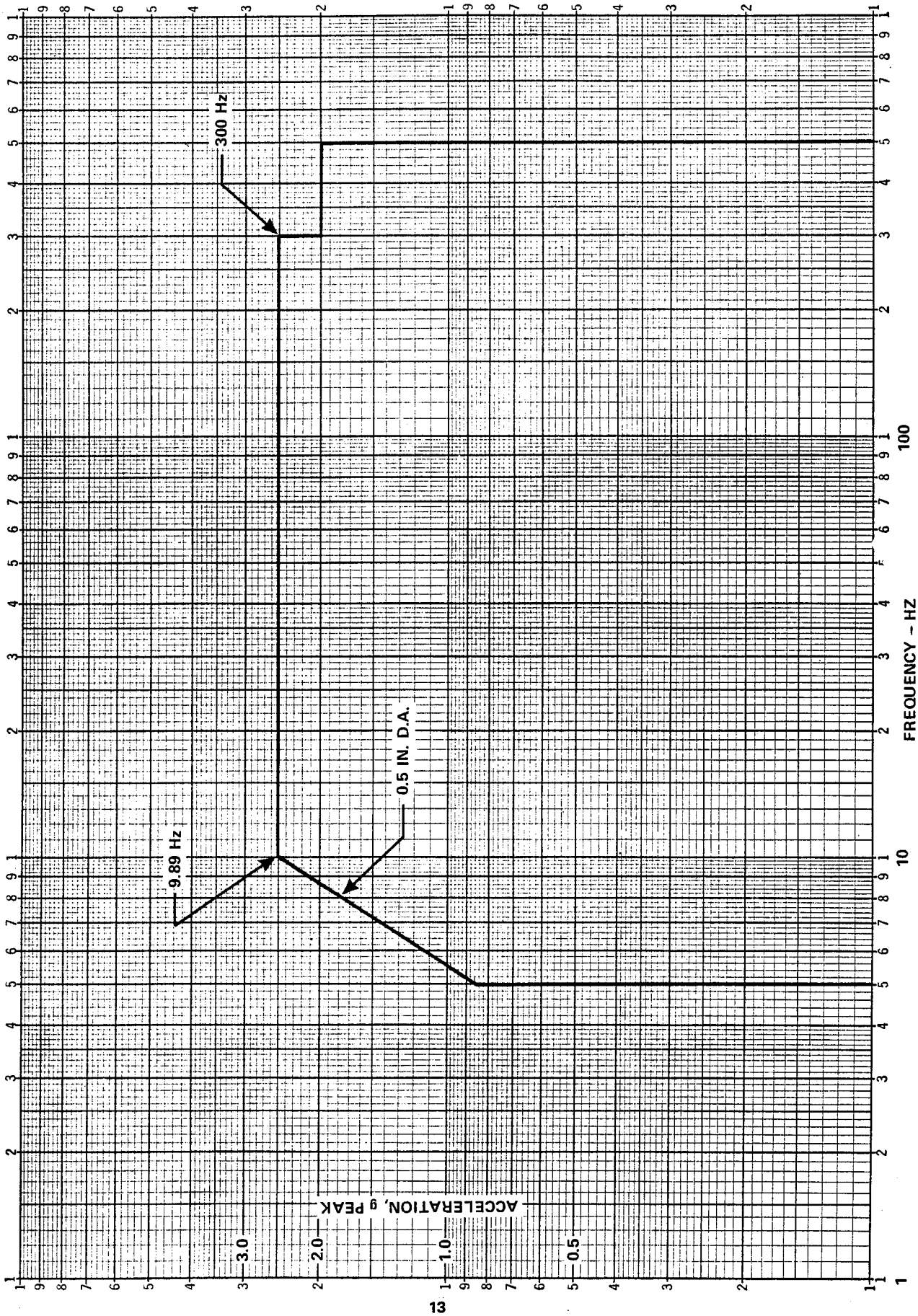


FIGURE 1. Vibration profile.

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4. QUALITY ASSURANCE PROVISIONS

SAFETY PRECAUTIONS

These limitations are recommended for the protection of personnel against potential hazards from backscattered laser radiation:

1. Conduct no test in rain, snow or hail.

2. Equip all personnel using the rangefinder and all personnel in the vicinity with laser safety glasses of at least density 4 at 6943 angstroms when testing under these conditions:

   a. Morning twilight
   b. Evening twilight
   c. Daytime heavy fog where visibility is less than 200 meters.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three rangefinders selected at random by the Government representative from the first fifteen production rangefinders.

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4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables V, VI and VII.

4.4.3 Failure. Failure of any rangefinder to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any rangefinder to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

TABLE V. First article tests.

Item	Characteristic	Requirement	Test Procedure
1	Sealing	3.5	4.6.4
2	Shock	3.4.1	4.6.3.1
3	Vibration A	3.4.2	4.6.3.2
4	High Temperature	3.4.3 and 3.4.4	4.6.3.3
5	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
6	Humidity	3.4.5	4.6.3.5
7	Altitude	3.4.6	4.6.3.6
8	Dust	3.4.7	4.6.3.7
9	Salt fog	3.4.8	4.6.3.8
10	Water hosing	3.4.9	4.6.3.9

4.5. Quality conformance inspection. Inspection plans A and B shall apply.

4.5.1 Inspection plan A.

4.5.1.1 General sample and tests. One rangefinder, as a control sample, shall be selected at random by the Government representative from each 100 rangefinders produced and shall be subjected to all the tests in table VII.

4.5.1.2 Environmental sample and tests. Two rangefinders, as a control sample, shall be selected at random by the Government representative from each 50 rangefinders produced or from each month's production, whichever occurs first. All the tests in table VIII shall be applied separately to each rangefinder in the sample.

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4.5.1.3 Acceptance. Where any one rangefinder of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.5.2 Inspection plan B.

4.5.2.1 Sample and tests. Each rangefinder in every lot shall be subjected to all the tests in table V and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.2.2 Acceptance. Where any one rangefinder fails to meet any specified requirement, the defective rangefinder shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.3 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.4 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11743101.

TABLE VI. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	Power turn-on condition	3.3.1	4.6.2.1
102	Ranging operation	3.3.2	4.6.2.2
103	Range return selector	3.3.3	4.6.2.3
104	RESET switch	3.3.4	4.6.2.4
105	BATTLE RANGE switch	3.3.5	4.6.2.5
106	Laser MODE switch	3.3.6	4.6.2.6
107	Ranging limits, accuracy, resolution and rate	3.3.7	4.6.2.7
108	Light dimming control	3.3.8	4.6.2.8
109	LIGHT switch	3.3.9	4.6.2.9
110	Peak output power	3.3.10	4.6.2.10
111	Input voltage variation	3.3.11	4.6.2.11
112	Power drain	3.3.12	4.6.2.12
113	Primary power transient	3.3.13	4.6.2.13
114	Transmitter test	3.3.14	4.6.2.14
115	Built-in test	3.3.15	4.6.2.15
116	Remote controlled operations	3.3.16	4.6.2.16
117	Sight system	3.3.17	4.6.2.17
118	Optical operability	3.3.18	4.6.2.18
119	Sight system and laser beam alignment	3.3.19	4.6.2.19

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TABLE VII. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General specification	3.1.2	MIL-F-13926 - Visual

TABLE VIII. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration B	3.4.2	4.6.3.2
305	High temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low temperature	3.4.3 and 3.4.4	4.6.3.4
307	Sealing	3.5	4.6.4

4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in table IX shall be used where specified to perform the required test.

TABLE IX. Test equipment.

Item	Part Number	Nomenclature
1	11750222	Test Set - Laser System Test
2	11750232	Cabinet, Laser - F.A.T.
3	11821748	Test Set - Immersion Leak, Purge and Fill
4	Target No. 1	12 - inch minimum diameter circle, 15 percent minimum diffuse reflectivity and located at 215 ± 5 meters from window of Receiver-Transmitter. The exact location of the target within this tolerance shall be known within ±1 meter.
5	Target No. 2	23-inch minimum diameter circle, 15 percent minimum diffuse reflectivity and loacted at the actual distance of Target No. 3A or 3B minus 20

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meters, from window of Receiver-Transmitter. The exact location of the target within this tolerance shall be known within ±1 meter.		
6	Target No. 3A	4 feet square minimum, 15 percent minimum diffuse reflectivity, and located at 490 ±15 meters from window of Receiver-Transmitter. The exact location of the target within this tolerance shall be known within ±1 meter.
7	Target No. 3B	4 feet square minimum, 15 percent minimum diffuse reflectivity, and located at 810 ± 15 meters from window of Receiver-Transmitter. The exact location of the target within this tolerance shall be known within ± 1 meter.
8	Target No. 4A	4 feet square minimum, 80 percent minimum diffuse reflectivity, and located at 490 ± 15 meters from window of Receiver-Transmitter. The exact location of the target within this tolerance shall be known within ± 1 meter.
9	Target No. 4B	4 feet square minimum, 80 percent minimum diffuse reflectivity, and located at 810 ± 15 meters from window of Receiver-Transmitter. The exact location of the target within this tolerance shall be known within ± 1 meter.

4.6.2 Performance tests.

4.6.2.1 Power turn-on condition. Use item 1 of table IX to determine compliance with 3.3.1.

4.6.2.2 Ranging operation. Use item 1 of table IX to determine compliance with 3.3.2.

4.6.2.3 Range return selector. Use item 1 of table IX to determine compliance with 3.3.3.

4.6.2.4 RESET switch. Use item 1 of table IX to determine compliance with 3.3.4.

4.6.2.5 BATTLE RANGE switch. Use item 1 of table IX to determine compliance with 3.3.5.

4.6.2.6 Laser MODE switch. Use item 1 of table IX to determine compliance with 3.3.6.

4.6.2.7 Ranging limits, accuracy, resolution and rate. Use items 1, 4, 5, 6, 7, 8 and 9 of table IX to determine compliance with 3.3.7.

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4.6.2.8 Light dimming control. Use item 1 of table IX to determine compliance with 3.3.8.

4.6.2.9 LIGHT switch. Use item 1 of table IX to determine compliance with 3.3.9.

4.6.2.10 Peak output power. Use item 1 of table IX to determine compliance with 3.3.10.

4.6.2.11 Input voltage variation. Use item 1, 6 and 7 of table IX to determine compliance with 3.3.11.

4.6.2.12 Power drain. Use item 1 of table IX to determine compliance with 3.3.12.

4.6.2.13 Primary power transient. Use item 1 of table IX to determine compliance with 3.3.13.

4.6.2.14 Transmitter test. Use item 1 of table IX to determine compliance with 3.3.14.

4.6.2.15 Built-in test. Use item 1 of table IX to determine compliance with 3.3.15.

4.6.2.16 Remote controlled operations. Use item 1 of table IX to determine compliance with 3.3.16.

4.6.2.17 Sight system. Use item 1 of table IX to determine compliance with 3.3.17.

4.6.2.18 Optical operability. Use item 1 of table IX to determine compliance with 3.3.18.

4.6.2.19 Sight system and laser beam alignment. Use item 1 of table IX to determine compliance with 3.3.19.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use item 2 of table IX and subject the rangefinder to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 $\pm$4 g with a time duration of 18 $\pm$3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the rangefinder to the performance tests of table VI.

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4.6.3.1.2 High intensity. Use item 2 of table IX and subject the rangefinder to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ±10.0 g with a time duration of 1.5 ±0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the rangefinder to the performance tests of table VI.

4.6.3.2 Vibration.

4.6.3.2.1 Vibration A. Use item 2 of table IX and subject the rangefinder to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figures 1 and 2 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the rangefinder to the performance tests of table VI.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Subject the rangefinder to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall 125°F. The performance tests of table VI shall be applied.

4.6.3.4 Low temperature. Subject the rangefinder to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table VI shall be applied.

4.6.3.5 Humidity. Place the rangefinder in a humidity chamber and subject it to the humidity test specified in MIL-STD-810, method 507.1, procedure IV. During steps 4, 6 and 8 of procedure IV, subject the rangefinder to the performance tests of 4.6.2. After exposure, subject the rangefinder to the performance tests of table VI.

4.6.3.6 Altitude. Place the rangefinder in an altitude chamber and subject it to the altitude test specified in MIL-STD-810, method 500.1, procedure I, except that:

a. First, the rangefinder in a non-operating condition, shall be exposed to an atmospheric pressure equivalent to 40000 feet above sea level for a period of not less than 1 hour.

b. Second, with the chamber pressure increased to an atmospheric equivalent to 10000 feet above sea level, subject the rangefinder to the performance tests of table VI.

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4.6.3.7 Dust. Subject the rangefinder to the dust test specified in MIL-STD-810, method 510.1, procedure I. At the conclusion of this test, subject the rangefinder to the performance tests of table VI.

4.6.3.8 Salt fog. Subject the rangefinder to the corrosion test specified in MIL-STD-810, method 509.1, procedure I. At the conclusion of this test, subject the rangefinder to the performance tests of table VI.

4.6.3.9 Water hosing. Subject all surfaces and joints of the rangefinder to water pressure of 50 ±5 psi for 0 ±5 seconds from a distance of 3 ±0.1 feet applied through a one inch nozzle. Wipe the rangefinder dry, remove the covers and inspect the inside for water intake. At the conclusion of this exposure, subject the rangefinder to the performance tests of table VI and the sealing test of 4.6.4.

4.6.4 Sealing. Pressurize the rangefinder with dry nitrogen to 5.0 ±1.0 psig and perform the leakage test. After determination of compliance with the sealing requirement, purge the optical and transmitter compartments with dry nitrogen at 12 ±1 psig for not less than 30 minutes and reduce the pressure 4.5 ±0.5 psig. Purge all other compartments with dry nitrogen at 1 psig for not less than 5 minutes and reduce the pressure to 0.0 + 0.2 -0.0 psig. Nitrogen shall be BB-N-411, type I, class 1, grade B with a dewpoint not greater than -25°F.

4.6.4.1 Test. Use item 3 of table IX, place the rangefinder in the following environment for not less than 2 hours and verify that the pressure drop is not greater than 0.2 psig.

a.	Temperature	73 ±2.5°F
b.	Relative humidity	50 percent ±5 percent
c.	Atmospheric pressure	28.5 +2.0 -3.0 in. Hg.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11743101. The level of protection shall be as specified in the procurement document. (See 4.5.4)

6. NOTES

6.1 Intended use. The AN/VVG-2 Laser Rangefinder system is intended for use in the primary fire control system for the M60A3 Tank. This is accomplished by transmitting a pulse of laser light, receiving reflected light from the target and converting the time from transmission to reception into range data in meters.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable levels of preservation, packaging and packing.

d. Applicable stock number.

e. Provisions for first article testing.

6.3 Previous revisions of MIL-R-48595A was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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